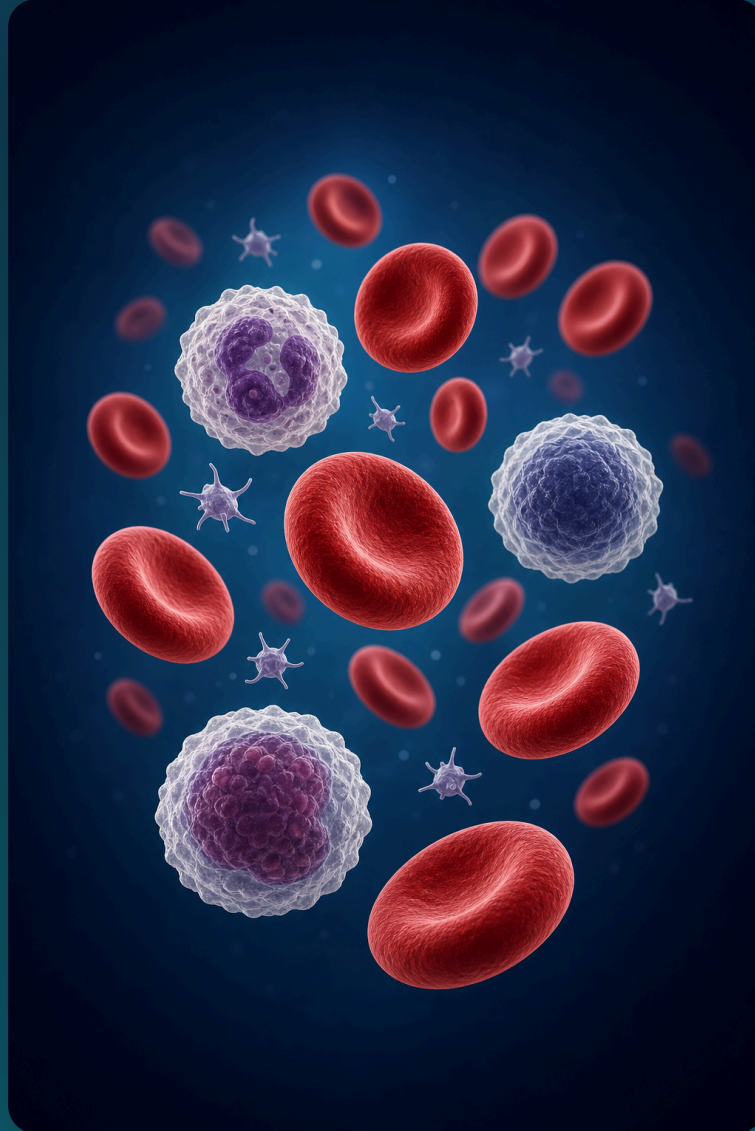


LABORATORY HEMATOLOGY



Essentials of Hematology

A Concise Illustrated Guide to Blood Cell Morphology & Coagulation

Visual atlas • English concise notes • For students & laboratory staff
Educational reference — not a substitute for validated clinical protocols.

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How to use this atlas

Part I (Morphology) trains the eye on what normal and abnormal blood cells look like. Part II (Coagulation) explains how the body forms and dissolves a clot, and how the lab measures it. Each topic is kept short: read the note, study the figure, then use the reference tables to revise.

1. How to Read a Peripheral Blood Smear

The blood film is the oldest and still one of the most informative tests in hematology. A well-made, well-stained smear lets you assess all three cell lines at once.

Making & examining the film

- **Preparation:** a small drop of EDTA blood is spread with a spreader slide at $\sim 30\text{--}45^\circ$, air-dried, then stained with a **Romanowsky** stain (Wright or Wright–Giemsa).
- **The reading zone:** examine where red cells *just* stop overlapping — the “monolayer”. Here central pallor and true morphology are visible.
- **Systematic scan:** low power (10 $\times$) for distribution & clumps $\rightarrow$ high dry / oil (40–100 $\times$) for detailed morphology $\rightarrow$ estimate a differential count.

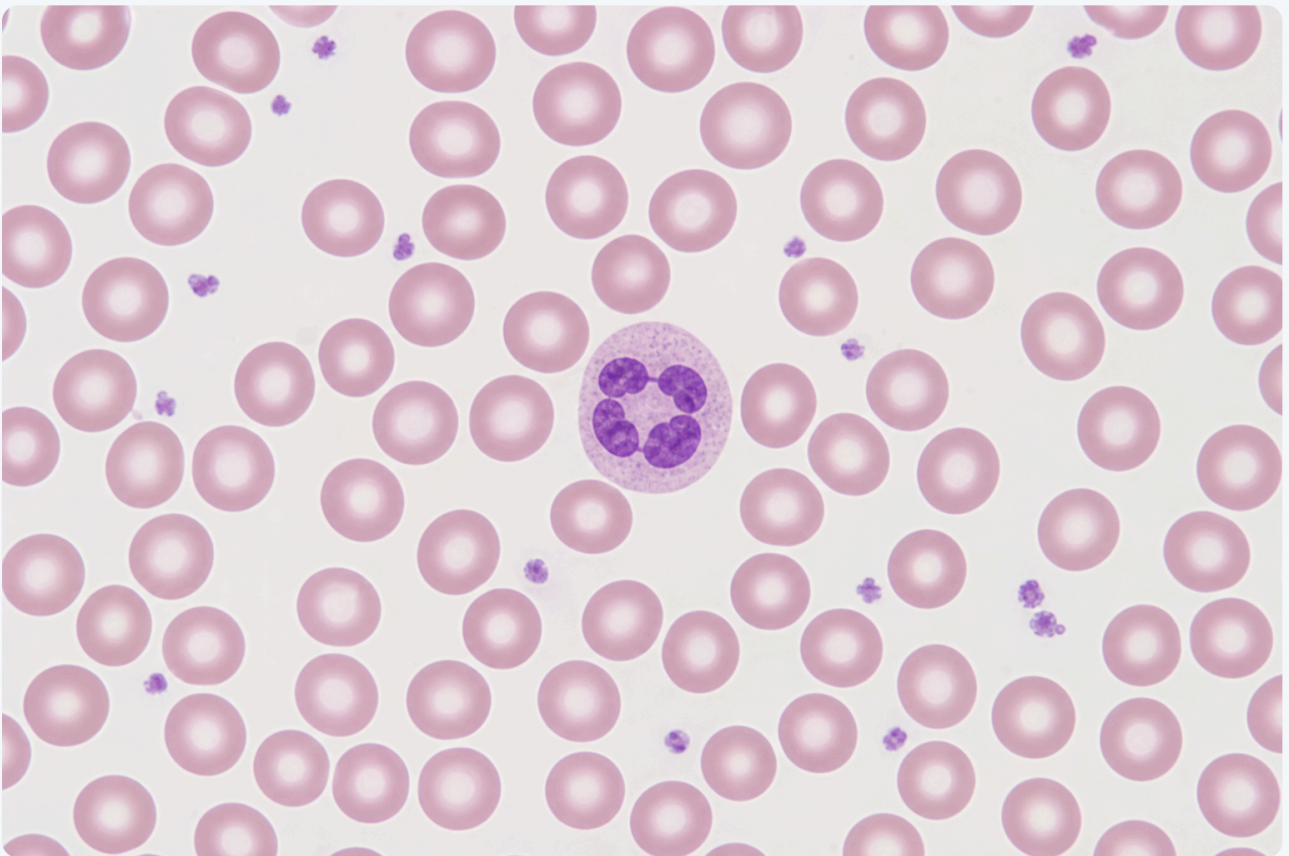


Fig 1.1 — Normal peripheral blood film. Uniform red cells with central pallor ($\sim 1/3$ of diameter), a segmented neutrophil, and scattered platelets. This is the reference against which all abnormalities are judged.

What to report

Red cells: size, colour, shape, inclusions. White cells: numbers, differential, maturity, reactive changes. Platelets: number (estimate), size, clumping. Always correlate with the automated CBC.

2. Red Blood Cell Morphology

RBC abnormalities are described in four axes: **size**, **colour**, **shape**, and **inclusions**. A normal RBC is 7–8 μm , biconcave, with central pallor.

Size (anisocytosis)

- **Microcyte** (<6 μm) — iron deficiency, thalassemia.
- **Macrocyte** (>9 μm) — B12/folate deficiency, liver disease, alcohol.
- **Anisocytosis** — variation in size (high RDW).

Colour (chromia)

- **Hypochromia** — enlarged pallor; low hemoglobin.
- **Polychromasia** — bluish young cells (reticulocytes); marrow response.

Shape (poikilocytosis)

- **Sickle cell** — HbS polymerization.
- **Target cell** — liver disease, thalassemia, HbC.
- **Spherocyte** — hereditary spherocytosis, immune hemolysis.
- **Schistocyte** — fragments; MAHA, DIC, mechanical valve.
- **Teardrop (dacrocyte)** — marrow fibrosis.
- **Elliptocyte** — hereditary elliptocytosis.

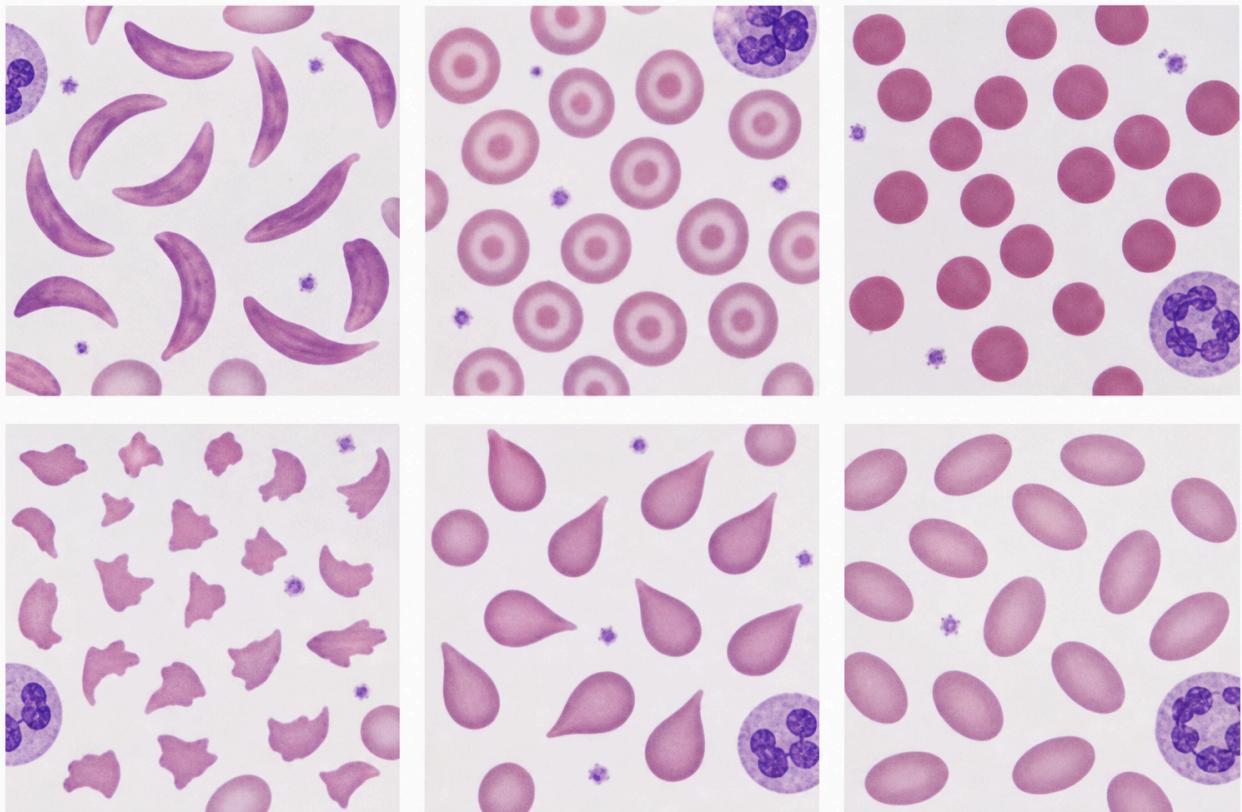


Fig 2.1 — Common poikilocytes. Top: sickle cells, target cells, spherocytes. Bottom: schistocytes (fragments), teardrop cells, elliptocytes. Recognising the dominant shape often points directly to a diagnosis.

RBC inclusions (brief)

Inclusion	Composition	Associated with
Howell–Jolly body	Nuclear (DNA) remnant	Post-splenectomy, hyposplenism
Basophilic stippling	Aggregated ribosomes	Lead poisoning, thalassemia
Pappenheimer bodies	Iron granules	Sideroblastic anemia, hyposplenism
Heinz body	Denatured hemoglobin	G6PD deficiency (supravital stain)

3. White Blood Cell Morphology

Five mature leukocytes circulate. Distinguish them by nuclear shape, cytoplasmic colour, and granules.



Fig 3.1 — The five leukocytes (left → right): neutrophil, lymphocyte, monocyte, eosinophil, basophil.

Cell	Approx. %	Key morphology	Main role
Neutrophil	40–70%	3–5 lobed nucleus, fine lilac granules	Bacterial phagocytosis
Lymphocyte	20–45%	Round dense nucleus, thin blue rim	Adaptive immunity (B/T)
Monocyte	2–10%	Large kidney-shaped nucleus, grey-blue cytoplasm, vacuoles	Macrophage precursor, antigen presentation
Eosinophil	1–6%	Bilobed nucleus, bright red-orange granules	Parasites, allergy
Basophil	<1%	Dark blue-black granules obscuring nucleus	Histamine, hypersensitivity

Left shift

Increase in immature neutrophils (band forms, metamyelocytes). Suggests acute infection or marrow stress.

Toxic changes

Toxic granulation, Döhle bodies, and vacuolation indicate severe infection/sepsis.

Reactive vs. neoplastic clues

- **Reactive lymphocytes** (e.g. EBV) — large, abundant pale-blue cytoplasm hugging surrounding red cells.
- **Blasts** — high nucleus:cytoplasm ratio, fine chromatin, nucleoli; Auer rods suggest AML. Always flag for review.

4. Platelet Morphology

Platelets are anucleate cytoplasmic fragments of megakaryocytes, 2–4 μm , with fine purple granules. Normal count 150–450 $\times 10^9/\text{L}$.

On the film

- **Number:** roughly 8–20 platelets per oil field $\approx$ normal; each platelet $\approx$ 15–20 $\times 10^9/\text{L}$.
- **Large / giant platelets:** ITP, myeloproliferative disease, inherited macrothrombocytopenias.
- **Platelet clumps:** EDTA-induced pseudothrombocytopenia — recollect in citrate.
- **Platelet satellitism:** platelets rosetting around neutrophils (EDTA artefact).

Always confirm low counts

A low automated platelet count must be checked against the film before reporting. Clumps, giant platelets, or fibrin strands can falsely lower the analyser count.

Critical value

Marked thrombocytopenia ($<20 \times 10^9/\text{L}$) carries a spontaneous bleeding risk and is usually a call-back result.

From morphology to function

Platelet number and size on the smear are the visual bridge to the next part of this atlas: platelets are the first responders of **hemostasis**. We now follow them into clot formation.

5. Overview of Hemostasis

Hemostasis is the balanced process that stops bleeding at a site of injury while keeping blood fluid everywhere else. It has three overlapping phases.

Phase	Main players	Result
1. Primary hemostasis	Platelets, von Willebrand factor, vessel wall	Unstable platelet plug (seconds–minutes)
2. Secondary hemostasis	Coagulation factors → thrombin → fibrin	Stable fibrin-reinforced clot
3. Fibrinolysis	Plasmin, tPA	Controlled clot breakdown & remodelling

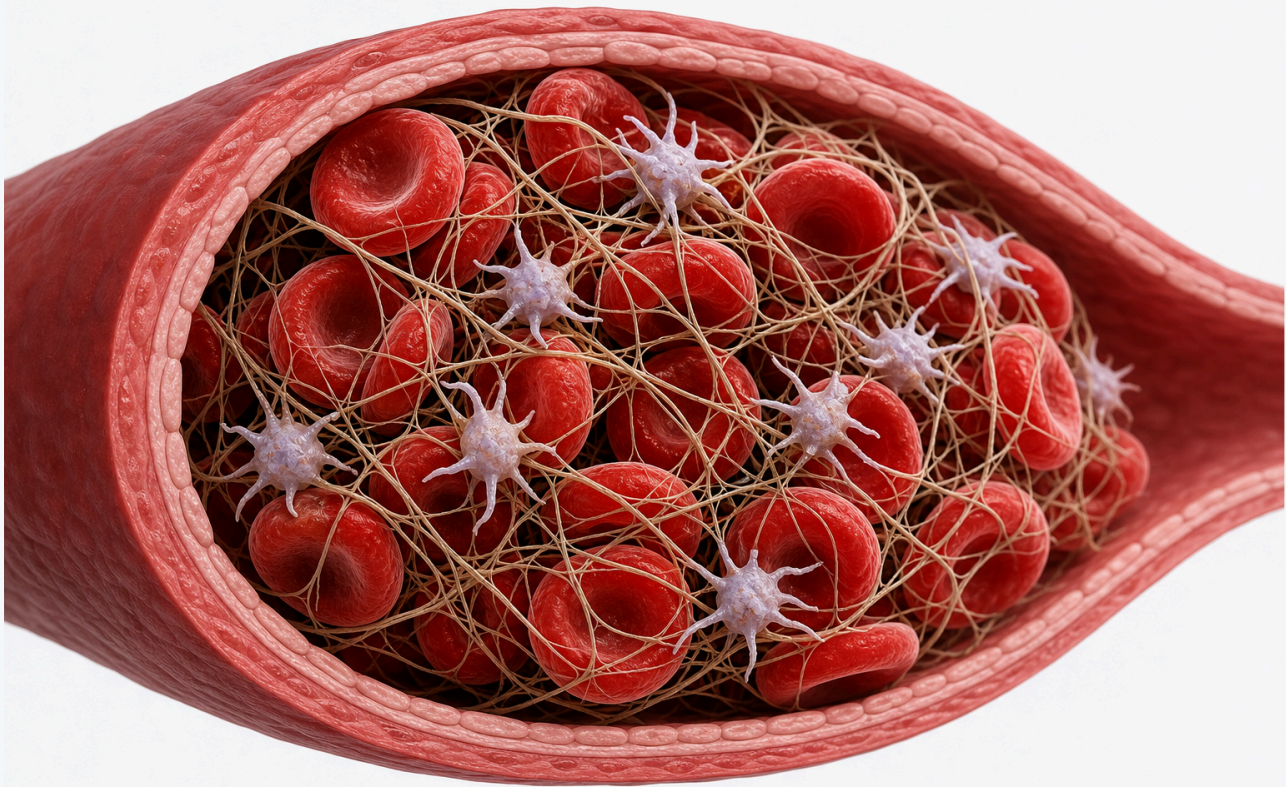


Fig 5.1 — The mature clot. A fibrin mesh (thin strands) cross-links activated platelets and traps red cells, converting the fragile platelet plug into a stable thrombus.

The balance concept

Procoagulant forces (platelets, clotting factors) are constantly opposed by anticoagulant and fibrinolytic forces (antithrombin, protein C/S, plasmin). Bleeding or thrombosis results when this balance is disturbed.

6. Primary Hemostasis — The Platelet Plug

Within seconds of vessel injury, platelets form the first seal in three steps: **adhesion** → **activation** → **aggregation**.

The three steps

- **Adhesion:** exposed subendothelial **collagen** binds **von Willebrand factor (vWF)**, which tethers platelets via receptor **GPIb**.
- **Activation:** platelets change shape, extend pseudopods, and release granule contents (**ADP**, thromboxane A_2 , serotonin), recruiting more platelets.
- **Aggregation:** activated **GPIIb/IIIa** receptors bind **fibrinogen**, cross-linking platelets into a plug.

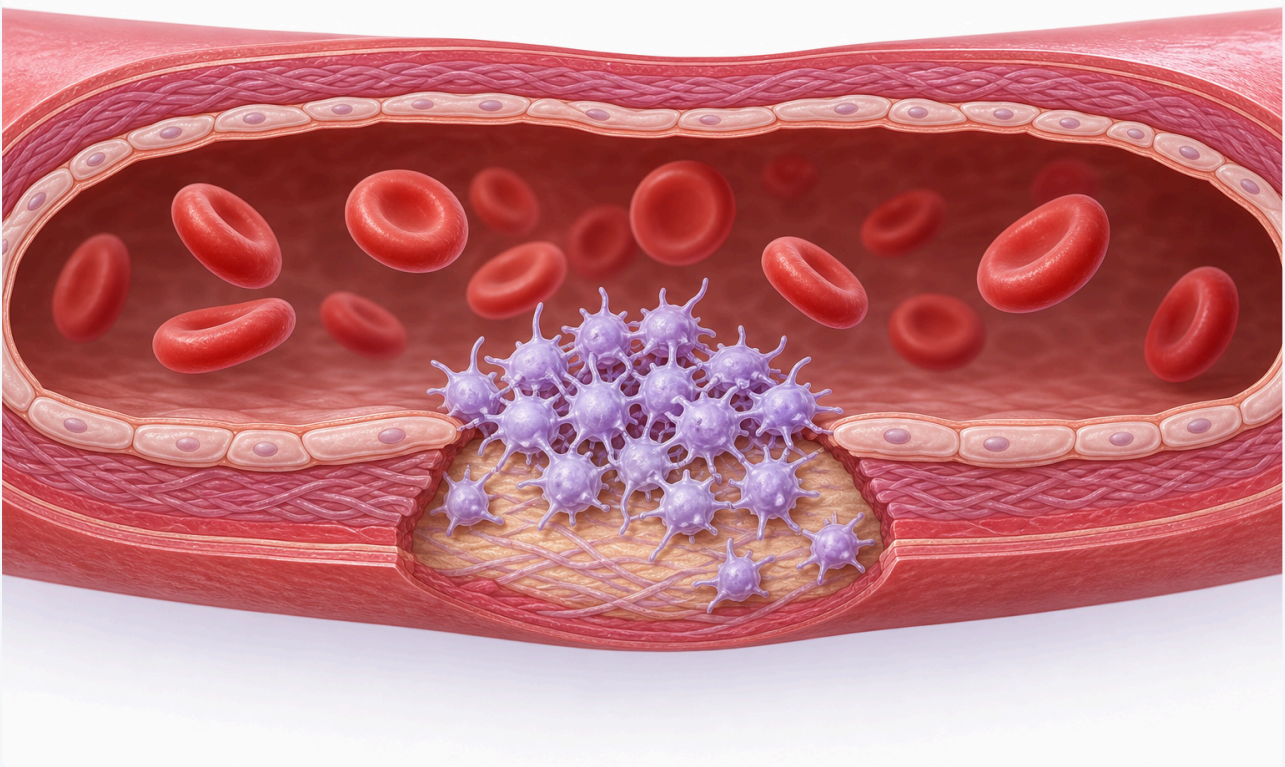


Fig 6.1 — Primary hemostasis. Platelets adhere to the exposed subendothelium at the site of injury and aggregate into a plug that temporarily seals the defect.

Disorders of primary hemostasis

Present with *mucocutaneous* bleeding: petechiae, easy bruising, epistaxis, menorrhagia. Examples: thrombocytopenia, von Willebrand disease, aspirin/clopidogrel effect, Glanzmann (GPIIb/IIIa), Bernard–Soulie (GPIb).

Test link

Screened with platelet count and platelet function assays (PFA-100/closure time); vWF activity for suspected VWD.

7. Secondary Hemostasis — The Coagulation Cascade

A chain of serine-protease reactions converts **fibrinogen** to **fibrin**. The classic model has an **intrinsic** and an **extrinsic** arm meeting in a **common** pathway. This model maps directly onto the lab tests.

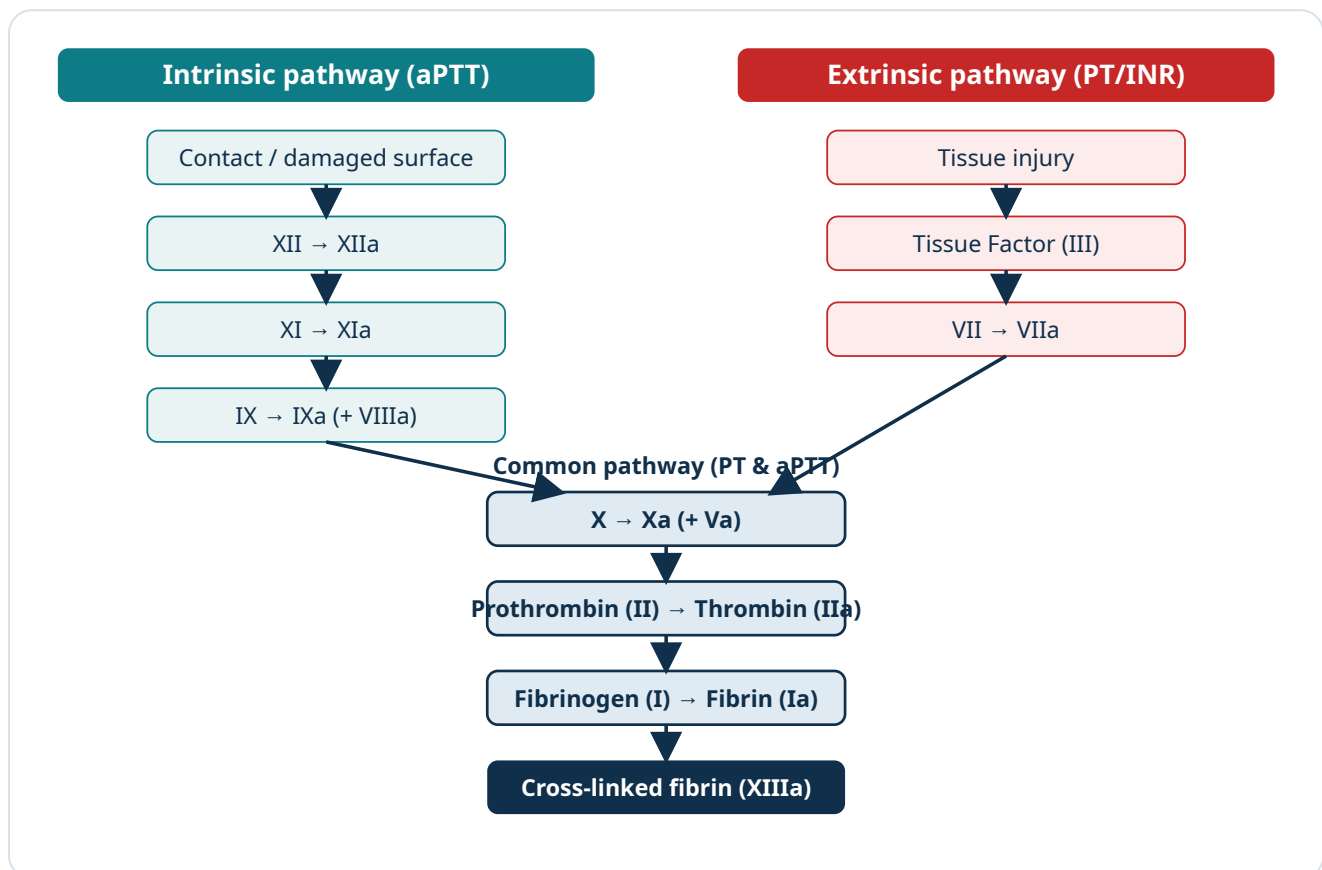


Fig 7.1 — Coagulation cascade. Factor numbering in Roman numerals; “a” = activated form. Modern *cell-based* models emphasise that tissue factor/VIIa initiates coagulation *in vivo*, with the intrinsic arm amplifying thrombin — but the classic diagram still explains PT and aPTT.

Vitamin K-dependent factors

II, VII, IX, X (and proteins C & S). Target of **warfarin**. Factor VII has the shortest half-life → PT rises first.

Cofactors & calcium

Factors V and VIII are cofactors (not enzymes). Ca^{2+} (factor IV) and phospholipid surfaces are required throughout.

8. Anticoagulants & Fibrinolysis

Clotting must be confined to the injury site. Natural inhibitors localise the clot, and fibrinolysis removes it once healing begins.

Natural anticoagulant systems

System	Action	Note
Antithrombin (AT)	Inhibits thrombin & Xa (IXa, XIa)	Accelerated by heparin
Protein C + Protein S	Inactivate factors Va & VIIIa	Activated by thrombin–thrombomodulin; vitamin K–dependent
Tissue factor pathway inhibitor (TFPI)	Inhibits TF–VIIa & Xa	Down-regulates initiation

Factor V Leiden

A mutation making factor Va resistant to protein C (“activated protein C resistance”) — the commonest inherited thrombophilia.

Fibrinolysis

- **Plasminogen** → **plasmin**, driven by **tissue plasminogen activator (tPA)** and urokinase.
- **Plasmin** digests fibrin, releasing **fibrin degradation products (FDPs)** and **D-dimer**.
- Regulated by **PAI-1** (inhibits tPA) and **α2-antiplasmin** (inhibits plasmin).
- **D-dimer** is a specific product of cross-linked fibrin breakdown — raised in DVT/PE, DIC, and many acute states.

9. Coagulation Laboratory Tests

Most bleeding/clotting screens rest on a handful of tests. Match the abnormal test to the pathway.

Test	Measures	Typical range*	Prolonged by
PT / INR	Extrinsic + common (VII, X, V, II, I)	PT 11–14 s; INR ~1.0	Warfarin, liver disease, factor VII deficiency
aPTT	Intrinsic + common (XII, XI, IX, VIII, X, V, II, I)	25–35 s	Heparin, hemophilia A/B, lupus anticoagulant
Thrombin time (TT)	Fibrinogen → fibrin (final step)	14–19 s	Heparin, low/abnormal fibrinogen
Fibrinogen	Factor I level	2–4 g/L	Low in DIC, liver disease
D-dimer	Fibrin breakdown	<0.5 mg/L FEU	Raised in VTE, DIC, inflammation
Platelet count	Primary hemostasis	150–450 ×10 ⁹ /L	—

*Reference ranges vary by reagent/analyser; always use your own laboratory's validated values.

Reading the pattern

- **PT ↑, aPTT normal** → extrinsic (factor VII), early warfarin.
- **aPTT ↑, PT normal** → intrinsic (VIII, IX, XI), heparin, lupus anticoagulant.
- **Both ↑** → common pathway, liver disease, DIC, high-dose anticoagulation.
- **Mixing study:** corrects → factor deficiency; fails to correct → inhibitor (e.g. lupus anticoagulant).

DIC pattern

↑PT, ↑aPTT, ↓platelets, ↓fibrinogen, ↑↑D-dimer, plus schistocytes on the film — a consumptive coagulopathy.

10. Quick Reference Tables

Morphology → Think of

Finding	First consider
Microcytic, hypochromic RBCs	Iron deficiency, thalassemia
Macrocytes + hypersegmented neutrophils	B12 / folate deficiency
Spherocytes	Hereditary spherocytosis, autoimmune hemolysis
Schistocytes + low platelets	MAHA / DIC / TTP-HUS
Target cells	Liver disease, thalassemia, HbC
Teardrop cells + leukoerythroblastic film	Marrow fibrosis / infiltration
Blasts / Auer rods	Acute leukemia — urgent review

Coagulation factors at a glance

Factor	Name	Pathway
I	Fibrinogen	Common
II	Prothrombin*	Common
III	Tissue factor	Extrinsic
IV	Calcium	All
V	Proaccelerin (cofactor)	Common
VII	Proconvertin*	Extrinsic
VIII	Antihemophilic (cofactor)	Intrinsic
IX	Christmas*	Intrinsic
X	Stuart-Prower*	Common
XI	PTA	Intrinsic
XII	Hageman	Intrinsic
XIII	Fibrin-stabilizing	Cross-linking

*Vitamin K-dependent (II, VII, IX, X). Hemophilia A = VIII deficiency; Hemophilia B = IX deficiency.

End of atlas

Morphology tells you *what* the cells look like; coagulation tells you *how* they stop bleeding. Together they cover the core of routine laboratory hematology.